

German Electricity Balancing & Imbalance Settlement

German Electricity Balancing & Imbalance Settlement

TSO Balancing Framework - DE-LU Control Area

Based on EU Electricity Balancing Guideline (EB GL) 2017/2195

1. Balancing Framework Overview

1.1 Responsible Parties

Entity	Role
Balance Responsible Party (BRP)	Responsible for balancing supply and demand in their portfolio
TSO (50Hertz, Amprion, TenneT DE, TransnetBW)	Maintains system balance, activates reserves
Balancing Service Provider (BSP)	Provides balancing energy (generators, demand response)
EPEX SPOT / exchanges	Provide intraday market for BRPs to self-balance

1.2 Balancing Products (Germany)

Product	Activation Time	Duration	Platform
FCR (Frequency Containment Reserve)	30 seconds	Continuous	PICASSO

Product	Activation Time	Duration	Platform
aFRR (Automatic Frequency Restoration)	5 minutes	15 min	PICASSO
mFRR (Manual Frequency Restoration)	12.5 minutes	15 min	MARI
RR (Replacement Reserve)	30 minutes	Variable	TERRE

2. Imbalance Settlement

2.1 Settlement Period

- Duration: 15 minutes (aligned with market time unit since Oct 2025)
- 96 settlement periods per day
- Each BRP settled independently per period

2.2 Imbalance Calculation

Imbalance (MW) = Actual_Generation + Actual_Imports – Actual_Demand – Actual_Exports – Scheduled_Position

Where Scheduled_Position = sum of all nominated trades (day-ahead + intraday)

2.3 Imbalance Pricing (Single Pricing - Germany)

Germany uses single imbalance pricing: Imbalance_Price = Marginal price of activated balancing energy in the settlement period

- If system is SHORT (under-generation): price = cost of upward balancing energy (high)
- If system is LONG (over-generation): price = cost of downward balancing energy (low/negative)
- BRPs with imbalance in SAME direction as system pay the imbalance price
- BRPs with imbalance in OPPOSITE direction receive the imbalance price

2.4 Imbalance Price Ranges (Typical)

System State	Imbalance Price Range	Frequency
Balanced (+/- 100 MW)	50-90 EUR/MWh	~60% of periods
Moderately short	80-150 EUR/MWh	~20% of periods
Moderately long	20-60 EUR/MWh	~15% of periods

System State	Imbalance Price Range	Frequency
Extremely short	150-9,999 EUR/MWh	~3% of periods
Extremely long	-500 to 20 EUR/MWh	~2% of periods

2.5 Imbalance Cost Calculation

$\text{Cost_to_BRP} = \text{Imbalance_Volume (MWh)} * (\text{Imbalance_Price} - \text{Reference_Price})$

Where Reference_Price = ID3 index (volume-weighted average of last 3h intraday trades)

3. Incentive Structure

3.1 Why Intraday Trading Reduces Imbalance Risk

- Intraday market gate closure: 5 min before delivery
- Imbalance settlement: after delivery
- Trading intraday allows BRP to adjust position based on latest forecasts
- Imbalance prices are MORE volatile than intraday prices (by design)
- Expected cost of being imbalanced > expected cost of trading intraday

3.2 Passive Balancing

- BRPs can intentionally deviate from schedule to help system balance
- If BRP deviates in OPPOSITE direction to system imbalance: receives favorable price
- This is LEGAL under REMIT (not manipulation) if based on public information
- Risk: if system direction changes, BRP pays unfavorable price

3.3 Imbalance Risk Quantification

$\text{Expected_Imbalance_Cost} = \text{P}(\text{short}) * \text{E}[\text{Imb_Price_Short} - \text{Market}] * \text{Volume}$
 $+ \text{P}(\text{long}) * \text{E}[\text{Market} - \text{Imb_Price_Long}] * \text{Volume}$

Typical values: - $\text{P}(\text{short})$ for wind portfolio: 30-40% of periods -
 $\text{E}[\text{Imb_Price_Short} - \text{Market}]$: 15-40 EUR/MWh (average penalty) - For 100 MW portfolio: expected imbalance cost = 5,000-15,000 EUR/day

4. Balancing Energy Platforms

4.1 PICASSO (aFRR)

- Platform for Automatic Frequency Restoration Reserve
- Activation: automatic, based on frequency deviation
- Merit order: cheapest bids activated first across all connected TSOs
- Settlement: pay-as-cleared (marginal pricing)
- Gate closure: 25 minutes before delivery

4.2 MARI (mFRR)

- Platform for Manual Frequency Restoration Reserve
- Activation: manual, by TSO operator
- Merit order: cheapest bids activated first
- Settlement: pay-as-cleared
- Gate closure: 25 minutes before delivery

4.3 Cross-Border Balancing

- Imbalance netting (IGCC): TSOs net opposing imbalances before activating reserves
 - Reduces total balancing costs by 200-400M EUR/year across Europe
 - Connected TSOs: 21 operational members across continental Europe
-

5. Data Sources for Imbalance Forecasting

5.1 Real-Time Data (Available to All Market Participants)

Source	Data	Update Frequency	URL
ENTSO-E Transparency	System balance, activated reserves	1 minute	transparency.entsoe.eu
TSO websites	Control area balance, frequency	10 seconds	50hertz.com, amprion.net
EPEX SPOT	Intraday prices, order book	Real-time	epexspot.com
Netztransparenz.de	Imbalance prices (ex-post)	15 minutes (T+10min)	netztransparenz.de

5.2 Forecasting Indicators

Indicator	Correlation with Imbalance Price	Source
Wind forecast error	HIGH (negative: over-forecast = long system)	ENTSO-E
Solar forecast error	MEDIUM	ENTSO-E
Temperature deviation	MEDIUM (cold snap = short system)	Weather services
Unplanned outages	HIGH (large outage = short system)	UMM platform
Cross-border flow changes	MEDIUM	ENTSO-E
Intraday price trend	HIGH (rising ID price = market expects shortage)	EPEX SPOT

6. Trading Strategy Implications

6.1 When to Trade Intraday vs Accept Imbalance

Trade intraday when: - Forecast error is large and direction is clear - Imbalance price spread (short vs long) is wide - Intraday liquidity is sufficient (volume > 50 MWh per QH) - Time to gate closure > 30 minutes (can still adjust)

Accept imbalance when: - Forecast uncertainty is symmetric (equal chance of long/short) - Intraday spread is already at imbalance price level - Volume is small (<5 MW) and transaction costs dominate - Gate closure is imminent and liquidity is thin

6.2 Optimal Position Management

1. After day-ahead: calculate expected position based on latest forecasts
2. At D-1 15:00 (intraday opens): place limit orders for known deviations
3. During delivery day: continuously update position as forecasts improve
4. Near gate closure: use market orders for remaining imbalance
5. Post-delivery: analyze imbalance costs and refine forecasting model